

MA 403 Real Analysis

Homework 10

1. Let $f : [a, b] \rightarrow \mathbb{R}$ be continuous, and suppose $\int_a^b f(x)dx = 0$. Prove that there exists a point $c \in [a, b]$ such that $f(c) = 0$.

Hint: Use Problem 7 from HW 9.

2. Let $f : [a, b] \rightarrow \mathbb{R}$ be continuous. Prove that there exists $c \in [a, b]$ such that

$$\int_a^b f(x)dx = (b - a)f(c).$$

Hint: Use Problem 5.

3. Suppose g is Riemann-integrable on $[a, b]$, and $f : [a, b] \rightarrow \mathbb{R}$ is a function such that $f(x) = g(x)$ except at a finite number of points x . Prove that f is Riemann-integrable and

$$\int_a^b f(x)dx = \int_a^b g(x)dx.$$

4. Let $f : [0, 1] \rightarrow \mathbb{R}$ be the function defined by

$$f(x) = \begin{cases} 0 & \text{if } x \text{ is irrational or zero} \\ \frac{1}{q} & \text{if } x = \frac{p}{q} \text{ in lowest terms.} \end{cases}$$

(Here the second description means: if $x \in \mathbb{Q}$, $x \neq 0$, and $x = \frac{p}{q}$ where $p, q \in \mathbb{Z}_{>0}$ such that p and q have no common factors other than 1.)

Prove that f is Riemann-integrable on $[0, 1]$ and has integral zero.

5. A function $f : [a, b] \rightarrow \mathbb{R}$ is called *piecewise-monotone* if there is a partition P of $[a, b]$, on each subinterval of which f is either increasing or decreasing. Prove that every piecewise monotone function is Riemann-integrable.
6. Give an example of a bounded function $f : [a, b] \rightarrow \mathbb{R}$ such that $|f|$ is Riemann-integrable but for which $\int_a^b f(x)dx$ does not exist.